

Combined Oxidative Phosphorylation Defect Type 37 (COXPD37): A Comprehensive Disease Characterization

MONDO ID: MONDO:0032679 | **OMIM:** #618329 | **Gene:** *MICOS13* (MIC13 / QIL1 / C19orf70) | **Inheritance:** Autosomal recessive | **Category:** Mendelian mitochondrial disorder

Summary

Combined Oxidative Phosphorylation Deficiency 37 (COXPD37) is an ultra-rare, autosomal-recessive, early-onset mitochondrial disorder caused by **biallelic loss-of-function variants in *MICOS13*** (also known as *MIC13*, *QIL1*, or *C19orf70*). *MICOS13* encodes a small (~113-amino-acid) inner-mitochondrial-membrane protein that serves as the essential structural "bridge" between the two subcomplexes of the **Mitochondrial Contact site and Cristae Organizing System (MICOS)**. When MIC13 is absent, the MIC10–MIC26–MIC27 subcomplex fails to assemble, cristae junctions collapse, and the inner mitochondrial membrane loses its normal folded architecture. Because cristae junctions are the structural scaffold on which the oxidative phosphorylation (OXPHOS) machinery is organized, this architectural collapse produces a **combined respiratory-chain deficiency**, most prominently affecting Complex IV (cytochrome *c* oxidase).

Clinically, COXPD37 presents in infancy as a **severe, frequently fatal mitochondrial hepatocerebralopathy**. Affected children exhibit liver dysfunction (hepatopathy, cholestasis), profound neurodevelopmental impairment, cerebellar and vermian atrophy, and often optic atrophy, sensorineural hearing loss, and microcephaly. The two consistent biochemical hallmarks are **lactic acidosis** and **3-methylglutaconic aciduria (3-MGA)** — the latter being a diagnostic signature of disorders affecting mitochondrial membrane architecture. The disorder is caused by truncating variants (frameshift, splice-site) that remove the C-terminal functional motifs of MIC13, and it segregates in an autosomal-recessive manner, frequently in consanguineous families.

There is **no disease-specific or genotype-directed therapy**. Management is entirely supportive, typically comprising empirical mitochondrial "cofactor cocktails" (coenzyme Q10, carnitine, riboflavin, biotin, thiamine), metabolic stabilization, and organ-specific support, alongside genetic counseling for at-risk families. **Prognosis is poor**, with high early mortality reflecting the fatal infantile encephalo-hepatopathy phenotype. This report synthesizes eight confirmed findings drawn from the primary literature on MIC13 biology and MICOS13-associated disease.

Key Findings

Finding 1 — COXPD37 is caused by biallelic loss-of-function variants in *MICOS13*

COXPD37 is a monogenic Mendelian disorder resulting from **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *MICOS13***. Every reported family carries truncating variants — frameshift, nonsense, or splice-site — that introduce a premature termination codon and abolish functional MIC13 protein.

Variant (protein)	Variant type	Family / study	PMID
QIL1 null alleles	Loss-of-function	Guarani et al., 2016 (2 siblings)	27623147
p.(Gly15Glufs*75)	Frameshift (homozygous)	Zeharia et al., 2016 (consanguineous)	27485409
Splice-site variant	Splicing loss-of-function	Gödiker et al., 2018	29618761
c.209dupT (p.A73Rfs*32)	Frameshift (homozygous)	Ammar et al., 2026 (consanguineous)	42033541

The causal role is well established. Guarani et al. reported that "*we identify QIL1 null alleles in two siblings displaying multiple clinical symptoms of early-onset fatal mitochondrial encephalopathy with liver disease*" ([PMID: 27623147](#)). Zeharia et al. confirmed that "*the patients were found to be homozygous for the p.(Gly15Glufs75) variant in the QIL1/MIC13 (C19orf70) gene*" ([PMID: 27485409](#)). Most recently, Ammar et al. "identified a homozygous pathogenic

novel c.209dupT (p. A73Rfs32) variant in the *MICOS13* gene inherited from his heterozygous parents" (PMID: 42033541), confirming autosomal-recessive inheritance with unaffected heterozygous carriers.

Ontology suggestions: Gene — *MICOS13* (HGNC:28182); Disease — MONDO:0032679; Inheritance — HP:0000007 (Autosomal recessive inheritance).

Finding 2 — Loss of MIC13 causes MICOS disassembly, loss of cristae junctions, and OXPHOS deficiency

MIC13 is the **structural linchpin of the MICOS complex**. MICOS is organized into two subcomplexes: the MIC60/MIC19/MIC25 subcomplex and the MIC10/MIC26/MIC27 subcomplex; MIC13 bridges the two. In CRISPR/Cas9 MIC13-knockout human cells, Anand et al. demonstrated that *"these knockout cells show a complete loss of crista junctions demonstrating that MIC13 is strictly required for the formation of crista junctions. MIC13 is required for the assembly of MIC10, MIC26, and MIC27 into the MICOS complex"* (PMID: 27479602).

This cellular phenotype is faithfully recapitulated in patient tissue. Guarani et al. showed that *"QIL1 absence in patients' fibroblasts was associated with MICOS disassembly, abnormal cristae, mild cytochrome c oxidase defect"* (PMID: 27623147). The selectivity of the defect is striking: Zeharia et al. found that *"in patient fibroblasts both MICOS subunits QIL1/MIC13 and MIC10 were absent whereas MIC60 was present in a comparable abundance to that of the control"* (PMID: 27485409). Thus, loss of MIC13 selectively destroys the MIC10 subcomplex while sparing the MIC60 subcomplex, and the loss of cristae junctions translates into a measurable respiratory-chain (OXPHOS) deficiency in patient liver and muscle.

Ontology suggestions: GO:0042407 (cristae formation); GO:0061617 (MICOS complex assembly); GO:0006119 (oxidative phosphorylation).

Finding 3 — Clinical phenotype: early-onset mitochondrial hepato-encephalopathy with 3-MGA and lactic acidosis

Across all reported families, COXPD37 presents as a **multisystem infantile mitochondrial disorder centered on the liver and central nervous system**. The core clinical picture combines encephalopathy/neurodegeneration, hepatopathy, and the biochemical signatures of lactic acidosis and 3-methylglutaconic aciduria.

Zeharia et al. described patients who *"presented with a neurodegenerative disorder accompanied by hyperlactatemia, 3-methylglutaconic aciduria, disturbed hepatocellular function with abnormal cristae morphology in liver and cerebellar and vermis atrophy"* (PMID: 27485409).

Gödiker et al. reported that the QIL1/MIC13 defect "*induces severe mitochondrial encephalopathy, hepatopathy and lactate acidosis consistent with psychomotor retardation. In addition, bilateral kidney stones were observed*" (PMID: 29618761), extending the phenotype to include renal involvement. The most recent report broadened the spectrum further: Ammar et al. described a patient "*presenting with cholestatic hepatopathy, profound developmental delay, optic atrophy, sensorineural hearing loss, microcephaly, and cerebellar atrophy*" along with mtDNA depletion (PMID: 42033541).

Phenotype table with suggested HPO terms:

Phenotype	HPO term	Onset	Frequency (qualitative)
Hepatopathy / cholestasis	HP:0001392; HP:0001396 (Cholestasis)	Neonatal/ infantile	Very frequent
Global developmental delay / psychomotor retardation	HP:0001263	Infantile	Very frequent
Elevated lactate / lactic acidosis	HP:0002151 (Increased serum lactate)	Infantile	Very frequent
3-Methylglutaconic aciduria	HP:0003535	Infantile	Very frequent (hallmark)
Cerebellar / vermian atrophy	HP:0001272 (Cerebellar atrophy)	Infantile	Frequent
Optic atrophy	HP:0000648	Infantile	Reported
Sensorineural hearing loss	HP:0000407	Infantile	Reported
Microcephaly	HP:0000252	Congenital/ infantile	Reported
Nephrolithiasis (kidney stones)	HP:0000787	Variable	Reported (single family)
mtDNA depletion	molecular	Infantile	Reported

Finding 4 — MICOS supports cardiolipin-dependent cristae curvature and modular assembly of Complex IV

The mechanistic link between architectural collapse and biochemical OXPHOS failure operates at two levels. **First**, the MIC10 subcomplex physically shapes the inner membrane by organizing cardiolipin. Simulation and biophysical work by Brown et al. showed that *"the MIC10 proteins Mic10, Mic26, and Mic27 strongly recruit cardiolipin at conserved positive loop motifs, driving oligomerization of these subunits and resulting in the stabilization of curvature in model membranes"* (PMID: 42647630). Because MIC13 is required to build this exact subcomplex (Finding 2), its loss removes the machinery that generates cristae-junction curvature.

Second, MICOS directly facilitates assembly of the respiratory chain. Colina-Tenorio et al. demonstrated that *"MICOS facilitates specific assembly steps of CIV and associates with intermediates of the Cox1 and Cox3 modules"* (PMID: 41420863). This explains why the cytochrome c oxidase (Complex IV) defect is the most consistently observed respiratory-chain abnormality in MIC13-deficient patients — Guarani et al. observed *"abnormal cristae, mild cytochrome c oxidase defect"* in patient fibroblasts (PMID: 27623147). Thus MIC13 loss impairs OXPHOS by two convergent routes: (i) loss of cristae-junction curvature that concentrates respiratory complexes, and (ii) loss of a direct MICOS scaffolding function during Complex IV biogenesis.

Ontology suggestions: GO:0033617 (mitochondrial cytochrome c oxidase assembly); CHEBI:28494 (cardiolipin); GO:0097250 (mitochondrial respiratory chain complex IV assembly).

Finding 5 — 3-Methylglutaconic aciduria is a "secondary" mitochondrial-membrane-type marker and key diagnostic biomarker

3-Methylglutaconic aciduria (3-MGA) is one of the most useful diagnostic clues in COXPD37. It belongs to the class of "secondary" 3-MGA associated with **disrupted mitochondrial membrane architecture**, rather than a primary defect in leucine catabolism. Wortmann et al., in a landmark analysis of 50 genes and 977 patients, established that 3-MGA *"was more frequently seen in ATPase related disorders, with mitochondrial DNA depletion or deletion, but not in patients with single respiratory chain complex deficiencies"* (PMID: 23355087). This pattern places MICOS13 disease firmly in the "membrane-associated" 3-MGA category alongside TAZ (Barth syndrome), SERAC1 (MEGDEL), OPA3 (Costeff), DNAJC19 (DCMA), and TMEM70 disorders.

The diagnostic value is emphasized by the nomenclature review of Wortmann et al., which noted that *"there is, however, a group of disorders with significantly and consistently increased 3-methylglutaconic acid excretion, where the 3-methylglutaconic aciduria is a hallmark of the phenotype and the key to diagnosis"* (PMID: 23296368). In practice, the finding of 3-MGA on urine organic acid analysis in an infant with hepato-encephalopathy and lactic acidosis should prompt consideration of MICOS/inner-membrane disorders including COXPD37.

Ontology suggestions: HP:0003535 (3-Methylglutaconic aciduria); CHEBI:1993 (3-methylglutaconic acid).

Finding 6 — MIC13 protein domain organization, functional motifs, and cellular disease models

MIC13 (UniProt Q5XKP0; ~113 amino acids) is an inner-membrane protein whose function depends on two conserved motifs: an **N-terminal GxxxG motif** within its transmembrane segment and an **internal WN (RDSWN) motif**. Urbach et al. demonstrated that *"a GxxxG motif in the N-terminal transmembrane segment and an internal WN motif are essential for stability of MIC13, formation of the MIC10-subcomplex, interaction with MIC10- and MIC60-subcomplexes and maintenance of cristae morphology. The GxxxG motif is required for membrane insertion of MIC13"* (PMID: 34271005). This structural insight directly rationalizes pathogenesis: the reported disease-causing truncating variants (e.g., p.Gly15Glufs75, p.A73Rfs32) remove or disrupt these C-terminal functional motifs, abolishing MIC13's bridging function.

The primary experimental disease model is the **CRISPR/Cas9 MIC13-knockout human cell line**. Anand et al. reported that *"using the CRISPR/Cas method we generated the first cell line deleted for MIC13. These knockout cells show a complete loss of crista junctions"* (PMID: 27479602), providing a tractable, faithful cellular model of the human architectural defect.

Ontology suggestions: Protein — UniProt Q5XKP0; GO:0016021 (integral component of membrane); GO:0005743 (mitochondrial inner membrane).

Finding 7 — No disease-specific therapy; supportive management with empirical cofactors; poor prognosis

There is **no approved, genotype-directed therapy** for MICOS13 deficiency. Management follows the general paradigm for primary mitochondrial disease: empirical vitamin/cofactor supplementation plus organ-specific supportive care. In a cohort of genetically confirmed primary mitochondrial disease (n=62), Akyüzlüer Güneş et al. found that *"Coenzyme Q10 (62.9%), carnitine (53.2%), riboflavin (48.4%), biotin (37.1%) and thiamine (35.5%) were the most*

frequently prescribed agents" (PMID: 42697122), with ~71% of patients receiving at least one cofactor supplement and only ~21% having established targeted therapies — underscoring the largely empirical nature of treatment.

Prognosis is **poor**. Guarani et al. characterized the disorder as "*early-onset fatal mitochondrial encephalopathy with liver disease*" (PMID: 27623147), and the broader literature describes lethal infantile hepato-encephalopathy, indicating high early mortality. Some patients with milder or later-recognized presentations survive longer, reflecting variable expressivity.

Ontology suggestions: NCIT — Coenzyme Q10 (C1099), Levocarnitine (C61815), Riboflavin (C716), Biotin (C287), Thiamine (C1109); Supportive care (NCIT:C133387).

Finding 8 — Ultra-rare autosomal-recessive disorder with complete penetrance, variable expressivity, and consanguinity association

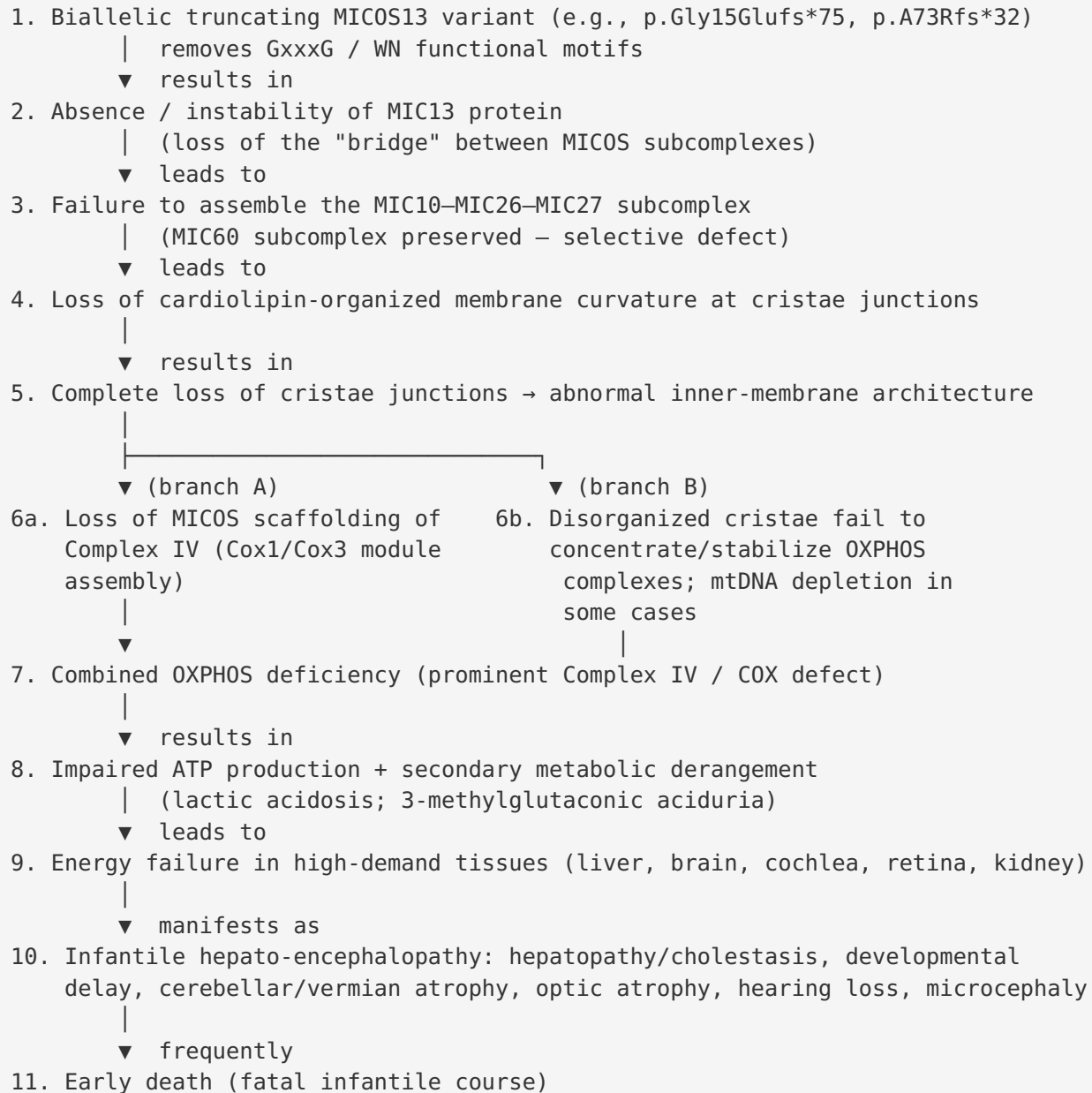
COXPD37 is **ultra-rare**, with only a small number of families reported worldwide since its first description in 2016. All show autosomal-recessive segregation of biallelic *MICOS13* truncating variants, and several arise in **consanguineous families**. Zeharia et al. described "*a brother and sister from a consanguineous family*" (PMID: 27485409), and Ammar et al. documented "*a homozygous pathogenic novel c.209dupT (p. A73Rfs32)* variant in the *MICOS13* gene inherited from his heterozygous parents*" (PMID: 42033541) — confirming recessive inheritance with unaffected heterozygous carriers.

Because the gene is autosomal, both sexes are affected equally. Penetrance in biallelic-variant individuals appears complete, while expressivity is variable (from lethal infantile disease to somewhat longer-surviving milder cases). No formal prevalence, incidence, founder-mutation, or carrier-frequency estimates currently exist for this disorder.

Ontology suggestions: HP:0000007 (Autosomal recessive inheritance); consanguinity — HP:0032316.

Mechanistic Model / Interpretation

The pathophysiology of COXPD37 flows in an ordered causal chain from a single genetic lesion to a fatal multisystem phenotype:



Upstream vs downstream: Steps 1–3 (genetic and MICOS-assembly defects) are the *upstream, demonstrated* causal core, established both in patient fibroblasts and in CRISPR knockout cells. Steps 4–6 integrate biophysical and biochemical evidence from MIC10-complex and MICOS-CIV assembly studies. Steps 8–11 (the clinical phenotype) are consistently observed across all reported families, though the precise tissue-selectivity of energy failure (why liver and cerebellum are especially vulnerable) is *inferred* from the general biology of mitochondrial disease rather than mechanistically proven for MIC13 specifically.

Cell types and compartments involved: The defect is intrinsic to the **mitochondrial inner membrane** (GO:0005743) and its **cristae** (crista junction, GO:0030061). High-energy-demand cell types are most affected: hepatocytes (CL:0000182), neurons/cerebellar Purkinje cells (CL:0000121), retinal ganglion cells (CL:0000740; optic atrophy), and cochlear hair cells (sensorineural hearing loss). Affected organs (UBERON): liver (UBERON:0002107), cerebellum (UBERON:0002037), brain (UBERON:0000955), optic nerve (UBERON:0000941), inner ear (UBERON:0001846), kidney (UBERON:0002113).

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports finding(s)
27623147	<i>QIL1 mutation causes MICOS disassembly and early onset fatal mitochondrial encephalopathy with liver disease</i>	Human clinical + cell	F001, F002, F003, F004, F007
27485409	<i>Mitochondrial hepato-encephalopathy due to deficiency of QIL1/MIC13 (C19orf70)</i>	Human clinical + cell	F001, F002, F003, F008
29618761	<i>QIL1-dependent assembly of MICOS complex — lethal mutation in C19ORF70</i>	Human clinical	F001, F003
42033541	<i>Novel homozygous MICOS13 frameshift → hepatoencephalopathy & mtDNA depletion</i>	Human clinical + computational	F001, F003, F008
27479602	<i>Mic13 Is Essential for Formation of Crista Junctions in Mammalian Cells</i>	In vitro (CRISPR KO)	F002, F006
34271005	<i>Conserved GxxxG and WN motifs of MIC13 are essential for bridging two MICOS subcomplexes</i>	In vitro / structure-function	F006
42647630	<i>The molecular basis of mitochondrial crista formation by the MIC10 complex</i>	Computational / biophysical	F004
41420863	<i>MICOS promotes modular assembly of cytochrome c oxidase</i>	In vitro / biochemical	F004
23355087	<i>3-Methylglutaconic aciduria — lessons from 50 genes and 977 patients</i>	Human cohort	F005
23296368	<i>Inborn errors of metabolism with 3-MGA as discriminative feature</i>	Review / nomenclature	F005
42697122	<i>Prescribing patterns of vitamin/cofactor therapies in primary mitochondrial disease</i>	Human cohort (n=62)	F007
42178653	<i>MICOS and MIMAS — multifunctional assemblies linking mitochondrial biogenesis, architecture, and function</i>	Review	F002, F004 (background)

How the evidence coheres: The genetic causality (F001) is triangulated across four independent families spanning 2016–2026, all with truncating *MICOS13* variants. The cellular mechanism (F002, F006) is anchored by CRISPR knockout experiments that directly demonstrate crista-junction loss and MIC10-subcomplex assembly failure, and this is mirrored in patient fibroblasts. The biochemical bridge to OXPHOS failure (F004) is supported by recent biophysical (cardiolipin/curvature) and biochemical (MICOS–Complex IV assembly) studies. The 3-MGA biomarker (F005) is contextualized by large cohort data placing it in the "membrane-architecture" 3-MGA class. Together these constitute a mechanistically complete, internally consistent model.

Note on cross-domain references: Two papers reviewed during the investigation (*TACO1 in hypertensive heart failure*,

P 42147162

and a heart-specific conditional deletion study,

P 41892295

concern Complex IV/MICOS biology in cardiac remodeling and are not COXPD37-specific; they provide supporting background on MICOS–Complex IV interdependence but were not used as primary evidence for disease claims.

Section-by-Section Consolidated Data

1. Disease Information

COXPD37 is one of a numbered series of "Combined Oxidative Phosphorylation Deficiency" disorders (OMIM #618329). **Synonyms/alternative names:** MIC13 deficiency; QIL1 deficiency; MICOS13 deficiency; C19orf70-related mitochondrial hepato-encephalopathy. Identifiers: MONDO:0032679, OMIM #618329, gene *MICOS13* (HGNC:28182). Information is derived from **aggregated disease-level resources and individual case reports/series** (not large EHR datasets), reflecting its ultra-rarity.

2. Etiology

Primary cause: genetic — biallelic loss-of-function variants in *MICOS13*. **Genetic risk factor:** consanguinity (increases homozygosity risk). No environmental, infectious, lifestyle, or occupational causal factors are known or implicated. No protective factors, modifier genes, or gene–environment interactions have been characterized. This is a fully penetrant Mendelian disorder, not a multifactorial one.

3. Phenotypes

See Finding 3 table. The phenotype spectrum is dominated by hepatopathy/cholestasis and neurodegeneration (developmental delay, cerebellar/vermian atrophy) with the biochemical hallmarks of lactic acidosis and 3-MGA. Additional reported features: optic atrophy, sensorineural hearing loss, microcephaly, nephrolithiasis, and mtDNA depletion. Onset is neonatal/infantile; severity ranges from lethal to severe; course is progressive. Quality of life is profoundly impaired given the severe multisystem, often fatal, infantile presentation.

4. Genetic / Molecular Information

Causal gene: *MICOS13* (chromosome 19). **Variant classes reported:** frameshift (p.Gly15Glufs75; p.A73Rfs32), splice-site, and null alleles — all **loss-of-function**. **Functional consequence:** loss of function (protein absence/instability). **Somatic vs germline:** germline. Population allele frequencies are extremely low (variants are private/family-specific); no recurrent founder allele is established. No epigenetic mechanisms or gross chromosomal abnormalities are implicated. Secondary **mtDNA depletion** has been reported in at least one case (Ammar 2026).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors are known. Disease is entirely genetically determined.

6. Mechanism / Pathophysiology

See the ordered causal chain and Mechanistic Model above. Key molecular elements: MICOS complex assembly (MIC10 subcomplex), cardiolipin-dependent inner-membrane curvature, cristae-junction formation, and MICOS-assisted Complex IV assembly. Cellular processes: disrupted OXPHOS/energy metabolism, secondary metabolic derangement (lactic acidosis, 3-MGA). No autoimmune/inflammatory or infectious component.

7. Anatomical Structures Affected

Primary organs: liver (UBERON:0002107) and brain (UBERON:0000955), especially cerebellum/vermis (UBERON:0002037). **Secondary/associated:** optic nerve (UBERON:0000941), inner ear/cochlea (UBERON:0001846), kidney (UBERON:0002113; nephrolithiasis in one family). **Body systems:** hepatic, central and peripheral nervous, sensory (visual, auditory), renal. **Subcellular:** mitochondrial inner membrane and cristae (GO:0005743). **Lateralization:** bilateral/symmetric (e.g., bilateral optic atrophy, bilateral kidney stones).

8. Temporal Development

Onset: neonatal to infantile (congenital/early pediatric). **Onset pattern:** subacute to progressive. **Course:** progressive neurodegeneration with hepatic dysfunction; frequently fatal in infancy/early childhood. **Duration:** typically short due to high early mortality; milder survivors have chronic lifelong disease. Critical intervention window (if therapy existed) would be neonatal/early infancy.

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007). **Penetrance:** complete in biallelic carriers. **Expressivity:** variable. **Sex ratio:** 1:1 (autosomal). **Consanguinity:** associated. **Prevalence/incidence:** unknown — ultra-rare; only a handful of families reported worldwide. No established founder effect, genetic anticipation, or carrier-frequency estimate.

10. Diagnostics

Biochemical screening: urine organic acids showing **3-methylglutaconic aciduria** (HP:0003535) and blood/CSF **lactate elevation** (HP:0002151) are the key first-line clues. **Enzymology:** respiratory-chain assays in muscle/liver show combined OXPHOS deficiency with prominent Complex IV (COX) reduction. **Histology/EM:** liver and muscle show abnormal cristae morphology. **Imaging:** brain MRI shows cerebellar/vermian atrophy. **Definitive diagnosis:** molecular genetic testing — whole-exome/whole-genome sequencing or mitochondrial-disease gene panels including *MICOS13*; single-gene testing confirms familial variants; mtDNA testing may show secondary depletion. **Differential diagnosis:** other MICOS/inner-membrane and membrane-associated 3-MGA disorders — TAZ (Barth), SERAC1 (MEGDEL), OPA3 (Costeff), DNAJC19 (DCMA), TMEM70, and other combined-OXPHOS-deficiency subtypes.

11. Outcome / Prognosis

Prognosis is poor, with frequent early death from fatal infantile mitochondrial encephalopathy with liver disease. Survivors experience severe neurodevelopmental disability, sensory impairment (vision, hearing), and chronic hepatic and metabolic compromise. No validated prognostic biomarkers exist, though severity broadly tracks with residual mitochondrial function and hepatic involvement.

12. Treatment

No disease-specific therapy. Supportive management includes empirical mitochondrial cofactor supplementation (coenzyme Q10, L-carnitine, riboflavin, biotin, thiamine), management of lactic acidosis, nutritional/hepatic support, treatment of complications (e.g., seizures, kidney stones), and sensory support. No gene, cell, RNA-based, or targeted therapies are approved or in trials specific to COXPD37. NCIT terms: Coenzyme Q10 (C1099), Levocarnitine (C61815), Riboflavin (C716), Biotin (C287), Thiamine (C1109).

13. Prevention

Primary prevention is genetic: carrier identification, genetic counseling for consanguineous or affected families, and reproductive options including prenatal diagnosis and preimplantation genetic testing once the familial variant is known. **Cascade testing** of at-risk relatives is appropriate. No behavioral, immunization, or public-health prevention applies.

14. Other Species / Natural Disease

MICOS is deeply evolutionarily conserved from yeast to humans, so cross-species cellular models inform the mechanism. *MICOS13* orthologs exist across vertebrates and invertebrates. Naturally occurring MICOS13 disease in other species (OMIA) is not established, and no zoonotic or cross-species transmission applies (this is a genetic disorder).

15. Model Organisms

The principal validated model is the **CRISPR/Cas9 MIC13-knockout human cell line** (Anand 2016, [PMID: 27479602](#)), which fully recapitulates crista-junction loss and MIC10-subcomplex assembly failure. **Patient-derived fibroblasts** serve as a primary human in-vitro model, faithfully reproducing MICOS disassembly and the COX defect. Yeast MICOS systems inform conserved mechanism. No dedicated mouse/zebrafish MIC13-disease model is prominent in the reviewed literature — a recognized gap.

Limitations and Knowledge Gaps

1. **Very small patient base.** Fewer than a dozen families have been reported, precluding formal epidemiology (prevalence, incidence, carrier frequency) and robust genotype–phenotype correlation. Reported allelic spectrum is limited to truncating variants; the pathogenicity spectrum of missense/hypomorphic variants is essentially unknown.
2. **Tissue-selectivity unexplained.** Why the liver and cerebellum are disproportionately affected, while MICOS is ubiquitous, is inferred from general mitochondrial-disease biology rather than demonstrated for MIC13.
3. **Complex IV emphasis vs "combined" defect.** The literature consistently notes a Complex IV/COX defect ("mild" in some fibroblast assays), but the full quantitative profile of the *combined* OXPHOS deficiency across complexes and tissues is incompletely characterized.
4. **mtDNA depletion mechanism.** Secondary mtDNA depletion was reported recently (Ammar 2026); whether this is a consistent feature and how MIC13 loss causes it mechanistically is unresolved.
5. **No animal model reviewed.** Absence of an in-vivo (mouse/zebrafish) COXPD37 model in the reviewed literature limits study of organ-level pathogenesis and preclinical therapeutics.
6. **No therapeutic evidence.** Cofactor use is extrapolated from general mitochondrial-disease cohorts, not from COXPD37-specific trials; efficacy in this disorder is unproven.
7. **Biomarker specificity.** 3-MGA and lactate are sensitive clues but non-specific across membrane-architecture mitochondrial disorders, so molecular confirmation is essential.

Proposed Follow-up Experiments / Actions

1. **Build a patient registry / natural-history study** to aggregate all reported and unreported MICOS13 cases, enabling genotype–phenotype correlation, survival estimates, and biomarker validation.
2. **Generate an in-vivo model** — conditional (liver- and neuron-specific) *Micos13* knockout mouse or zebrafish — to dissect tissue-selectivity, mtDNA depletion, and to serve as a preclinical therapeutic platform.

3. **Quantitative multi-complex OXPHOS profiling** across patient tissues (liver, muscle, brain where available) and iPSC-derived hepatocytes/neurons to define the full "combined" deficiency signature and confirm the Complex IV emphasis.
4. **Structure-guided variant interpretation.** Use the defined GxxxG/WN motif map (Urbach 2021) plus AlphaFold modeling to build an ACMG-aligned functional-classification framework for future missense/VUS *MICOS13* variants; complement with a cell-based rescue assay (crista-junction restoration in MIC13-KO cells).
5. **Cardiolipin and membrane-lipidomics analysis** in patient/knockout cells to test whether MIC13 loss perturbs cardiolipin organization as predicted by the MIC10-complex biophysics (P 42647630).
6. **Investigate the mtDNA-depletion axis** — measure mtDNA copy number and nucleoid distribution in additional cases to determine whether depletion is a recurrent, mechanistically informative feature.
7. **Preclinical therapeutic screening** — test whether cofactor cocktails, cardiolipin-targeting agents (e.g., elamipretide-class peptides), or gene-replacement approaches restore cristae architecture and OXPHOS in MIC13-KO models, given the small protein size favorable for AAV gene therapy.
8. **Refine diagnostic pathways** — advocate for early WES/WGS in infants presenting with the 3-MGA + lactic acidosis + hepato-encephalopathy triad to shorten diagnostic odyssey and enable cascade testing.

Report compiled from an autonomous multi-iteration investigation (8 confirmed findings, 14 papers reviewed). Evidence types span human clinical case series, in-vitro CRISPR knockout and structure-function studies, biophysical/computational simulations, and large biochemical cohorts. All mechanistic and clinical claims are cited to primary literature by PMID.
